Quantum-Tube 'Prime Resonator' Presentation

1. Concept Overview

Create a planar array of thin waveguides ('tubes') that implement arithmetic translations T■ as unitary propagation-and-coupling
Each prime p gets a dedicated guide section of physical length ■■ = $\alpha \cdot \log p$, with couplers routing amplitude from node n to no
The one-step propagator reproduces Hδ(s) = ∑■ (p■■ T■ + p■(1–■) T■*), with weights realized by phase shifters and tap cou

2. System Architecture (Photonic SiN PIC)

Substrate: 200 mm Si wafer with 4–6 µm thermal SiO■ bottom cladding.
Core: LPCVD SiN, 300 nm thickness (single-mode at 1550 nm with 1.2 µm width).
Top cladding: PECVD SiO■, 2–3 µm.
Prime waveguide bank {W■}: straight sections with lengths ■■ = α·log p.
Coupler matrix: low-loss couplers for forward T■ and reverse T■* channels.
Phase/attenuation weights: thermo-optic phase shifters impose ϕ■ and gains.
Node lattice: finite set N = {1,…,N} with routing from node n to pn.
Scaling constant $\alpha \approx 4.95$ mm for τ■ = 20 ps and n_eff ≈ 1.75.

3. Concrete Device Spec (Rev-A Demo)

Die size: 35 mm × 25 mm (SiN PIC).
Prime set: $p \in \{2,3,5,7,11,13,17,19,23,29\}$.
Node count: $N = 16$.
Waveguide geometry: 300 nm × 1.2 µm; $n_{eff} \approx 1.75$.
Propagation loss: ≤ 0.2 dB/cm.
Couplers: 3 dB MMI + variable directional couplers with thermo-optic tuners.
Phase shifters: 200 µm heaters; 2π phase at ≈ 25 mW.
I/O: 8 grating couplers for inputs; 16 for outputs.

4. Measurement Protocol (ζ-line Mapping)

Calibration: $\sigma = 1/2$, zero phases ($t=0$), equalize path losses.
Critical-line sweep: fix $\sigma = 1/2$, ramp global phase K to sweep t.
Record $H(t)$: reference input port and composite output.
Zero signatures: deep minima in $|H(t)|$ align with ζ-zeros.
Symmetry check: swap forward/reverse banks to confirm mirrored response.

5. Error/Tolerance Budget

Length error Δ■■ ±5 µm → phase error Δφ■ ≈ ±0.035π.
Index drift Δn ±0.01 → uniform phase drift.
Coupler ratio error ±2%.
Loss ≤ 0.2 dB/cm → notch shallowing.
Thermal crosstalk –20 dB → phase cross-couple.
Laser λ drift ±0.1 nm → phase drift.

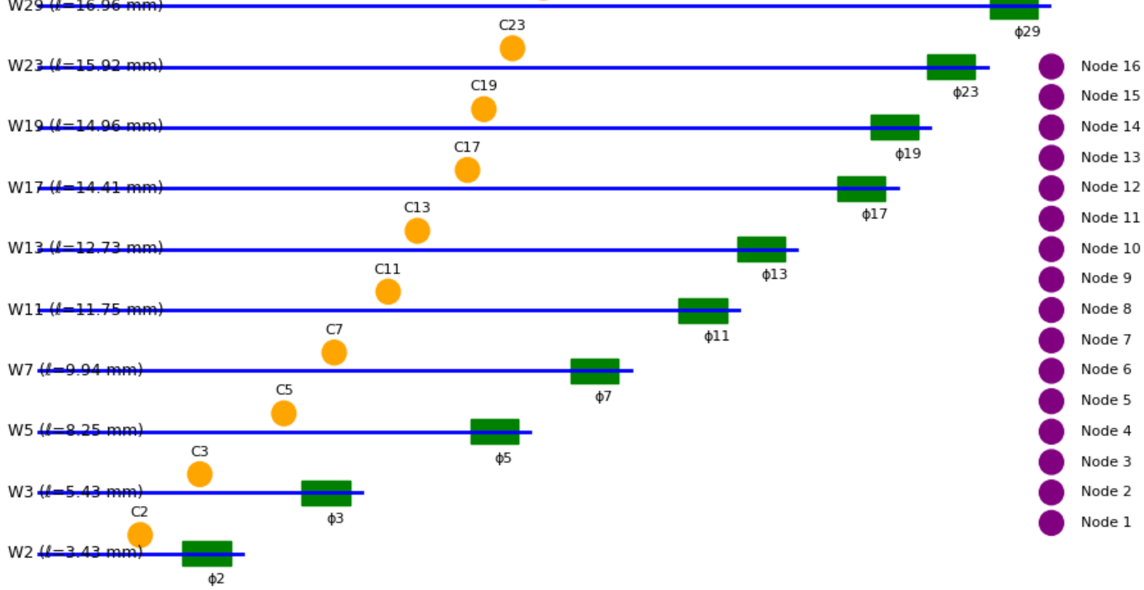
6. Provisional Patent Claim Map

Ind. Claim 1: Guided-wave device with paths of length ∝ log p and couplers/phasers implementing ∑■ (p■■ T■ + p■(1–■) T■

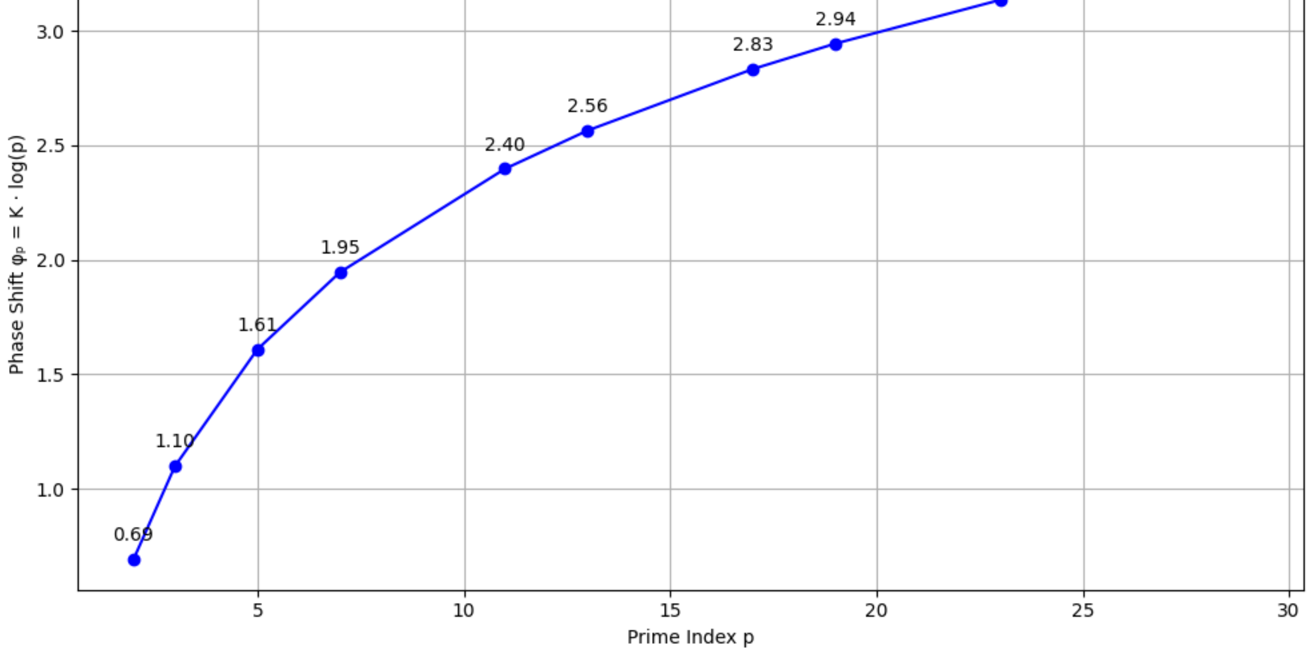
Ind. Claim 2: Global phase control ϕ■ = K·log p to scan spectral response.

Dep. Claims: SiN/SiO■ platform at 1550 nm; adjoint symmetry measurement; finite-P truncation; acoustic waveguide extension

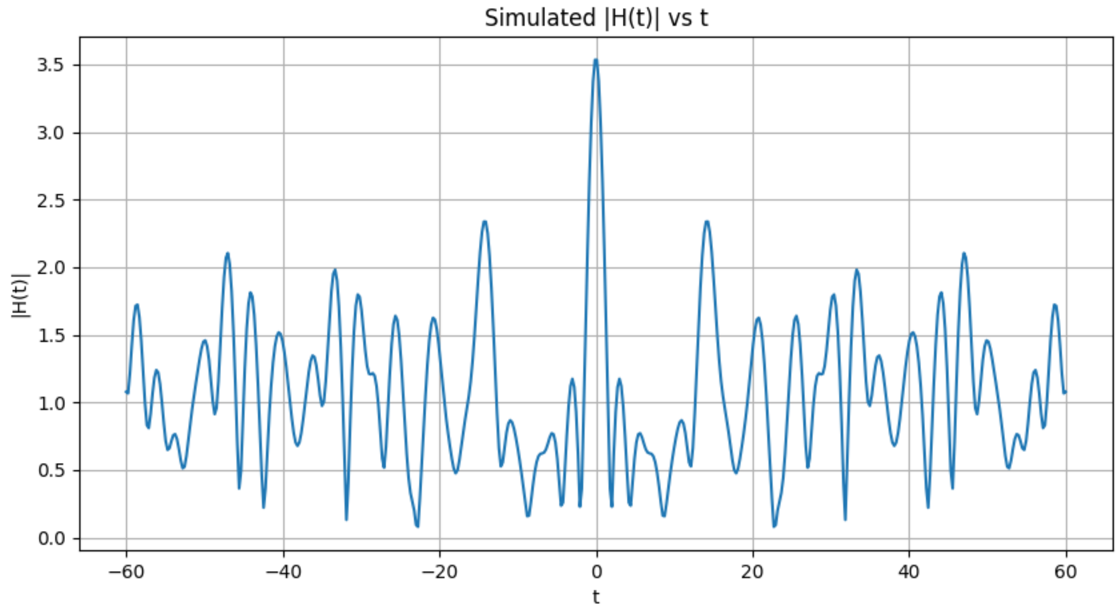
Trade-secret: heater calibration matrix, loss-equalization routine, routing compiler, optimal α,P,N scaling law.

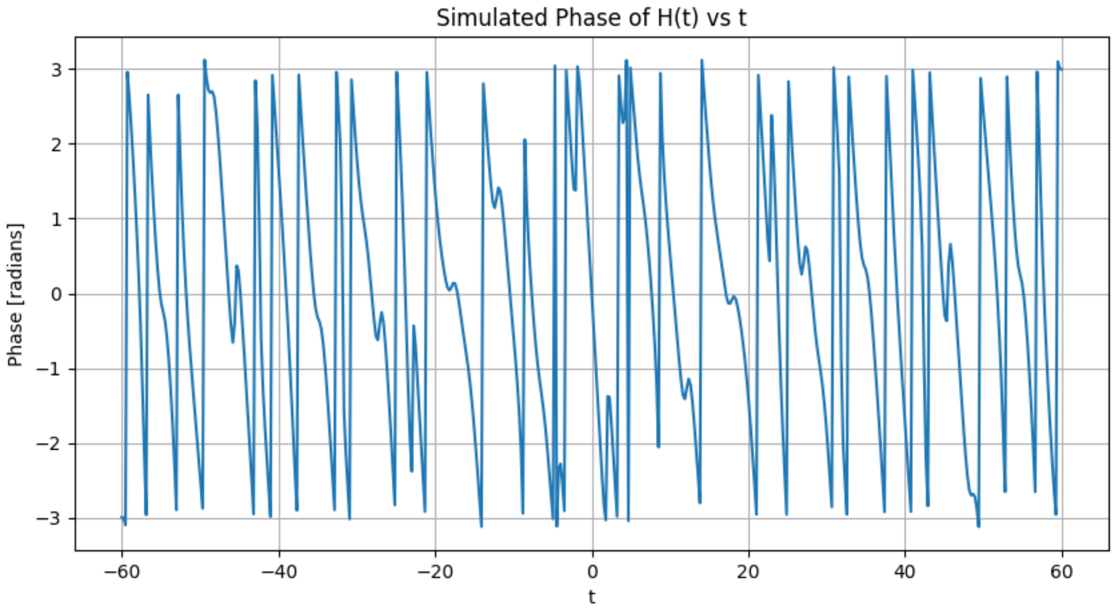


W29 (ℓ=16.96 mm)
ϕ29
C23
W23 (ℓ=15.92 mm)
ϕ23
C19
W19 (ℓ=14.96 mm)
ϕ19
C17
W17 (ℓ=14.41 mm)
ϕ17
C13
W13 (ℓ=12.73 mm)
ϕ13
C11
W11 (ℓ=11.75 mm)
ϕ11
C7
W7 (ℓ=9.94 mm)
ϕ7
C5
W5 (ℓ=8.25 mm)
ϕ5
C3
W3 (ℓ=5.43 mm)
ϕ3
C2
W2 (ℓ=3.43 mm)
ϕ2
Node 16
Node 15
Node 14
Node 13
Node 12
Node 11
Node 10
Node 9
Node 8
Node 7
Node 6
Node 5
Node 4
Node 3
Node 2
Node 1



0.69
1.10
1.61
1.95
2.40
2.56
2.83
2.94
3.0
2.5
2.0
1.5
1.0
5
10
15
20
25
30
Phase Shift φp = K · log(p)
Prime Index p


Simulated |H(t)| vs t
|H(t)|
t
3.5
3.0
2.5
2.0
1.5
1.0
0.5
0.0
−60
−40
−20
0
20
40
60


Simulated Phase of H(t) vs t
Phase [radians]
t
3
2
1
0
−1
−2
−3
−60
−40
−20
0
20
40
60

6·· Fabrication Process Pack (SiN Photonic Version)

A. Layer Stack (CMOS-Compatible)

Layer	Material	Thickness	Function
Top cladding	PECVD SiO·	2.5 µm	Optical isolation, mechanical passivation
Core	LPCVD SiN	300 nm	Single-mode waveguide (1550 nm)
Bottom cladding	Thermal SiO·	5 µm	Optical isolation
Substrate	Si	725 µm	Handle wafer
Heaters	TiN	100 nm	Thermo-optic phase shifters
Vias / pads	AlCu	1 µm	Routing to bond pads

Process:
- Grow 5 µm thermal oxide on Si.
- Deposit 300 nm LPCVD SiN; pattern with DUV stepper (248 nm).
- Etch (ICP-RIE, CHF·/O·).
- Deposit 2.5 µm PECVD SiO· top cladding.
- CMP for planarization.
- Deposit TiN, pattern heaters.
- Deposit AlCu pads, passivate, dice.

Yield target > 90%; propagation loss · 0.2 dB/cm; heater cross-talk < ·20 dB.

B. Mask Layers (GDS Tags)
- WG_CORE
- WG_OPEN (etch)
- HEATER
- PAD
- METAL_ROUTE
- OPEN_PASS
- ALIGN / MARK

C. Electrical Routing
- 16 heater lines (phase trims per prime path)
- 10 VDC (variable coupler) lines (gain p^··)
- 1 global ·t-dial· DAC
- SPI or I^2C driver ASIC or external micro-controller (STM32 class)

D. Packaging
- Butterfly-type with 24-pin header (20 control, 2 GND, 2 ref)
- Fiber array 8-in / 8-out (127 µm pitch, epoxy underfill)
- Thermal pad (Cu block) under die

7·· Acoustic Variant (Low-Cost Bench Prototype)

A. Core Geometry
- Medium: Air (v · 343 m/s)
- Tubes: Aluminum or 3D-printed PLA, internal Ø = 20 mm, wall 2 mm
- Lengths: L· = · log p, choose · = 0.33 m so L· = 0.23 m
  · L· = 0.36 m, L· = 0.55 m, L· = 0.66 m, L·· = 0.78 m, ·
- Couplers: T-junction manifolds with 3-way power split (forward/back/absorber)
- Attenuators: Acoustic foam inserts setting amplitude · p^··
- Phase control: Stub tuners (sliding pistons or micrometer screws)

B. Source & Readout
- Source: Swept sine 1·5 kHz loudspeaker
- Mic array (MEMS, 8 ch) at outputs
- Data: Transfer magnitude vs. frequency · compare to |·_P(· + it)| for t · frequency

C. Build Method
- 3D-print modular tubes, push-fit onto manifold plate
- Use microphone multiplexer + Teensy 4.1 for 8-ch FFT
- Software: Python + scipy.signal for spectral mapping

Result: Audible ·prime resonator· demonstration · at certain frequencies, near-total destructive interference

8·· Operator-to-Device Math Appendix

A. Discrete Operator
Your finite-prime compact operator:
K_P(s) = ·_{p · P} (p^{-s} T_p + p^{-(1·s)} T_p^*)
acts on a Hilbert space of sequences (or continuous fields along tubes).

B. Physical Realization
In the photonic or acoustic array:
- T_p · Unitary propagation of amplitude through a guide of delay ·_p = (·_p n_eff)/c
- Multiplication by p^{-s} · Attenuation g_p = p^{-·} and phase ·_p = ·t log p

Hence the network·s transfer function at complex ·frequency· s:
H_P(s) = ·_{p · P} [p^{-s} e^{i··_p} + p^{-(1·s)} e^{-i··_p}]

C. Determinant Identity
The measured overall transmission · det(I · K_P(s))^·1;
Zeros of · correspond to points where eigenvalue 1 of K_P(s) is reached (resonance · deep transmission no

D. Spectral Barrier
Tuning · shifts the average loop gain:
- · > ½ · Loop gain < 1 · Sub-resonant (smooth)
- · = ½ · Unitary edge · Notches appear
- · < ½ · Oscillatory divergence

E. Physical Scaling
·_p · log p ensures uniform sampling of log-frequency space:
Scanning frequency or phase dial corresponds to moving along Im s = t

Thus your photonic/acoustic device is a ·-analyzer in real space.